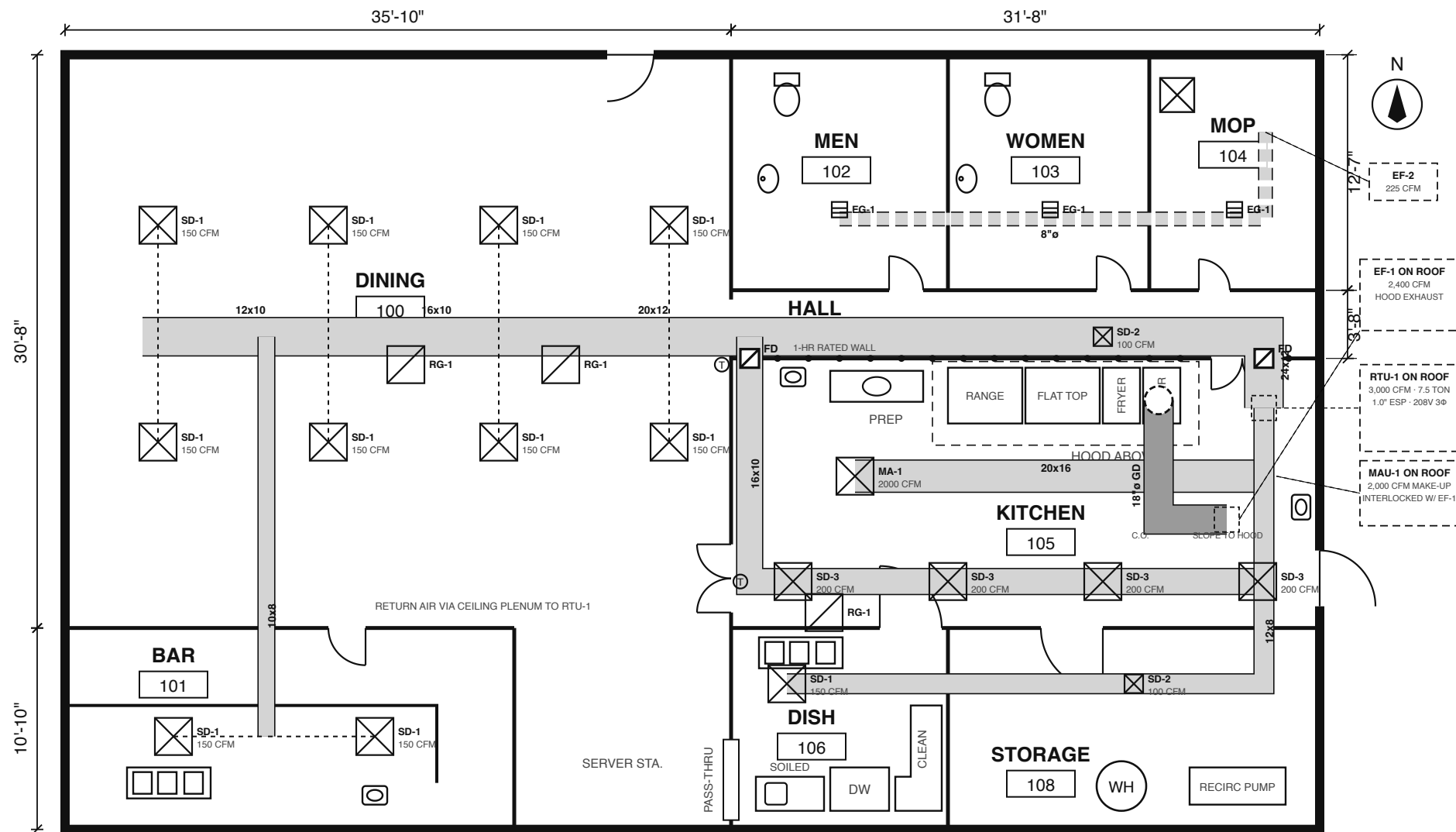


LEGEND

- SUPPLY DIFFUSER, TYPE AND CFM AS NOTED
- RETURN GRILLE, TO CEILING PLENUM
- EXHAUST GRILLE
- SUPPLY DUCT, SIZE AS NOTED, 1" W.G.
- EXHAUST DUCT
- GREASE DUCT (GD), WELDED 16 GA BLACK STEEL
- FLEX DUCT, 8"ø, 6'-0" MAX
- 1-HR RATED WALL · FD: FIRE DAMPER, UL 555

MECHANICAL KEYNOTES

SUPPLY DUCT: GALVANIZED, SMACNA
1" W.G., 2" EXTERNAL WRAP IN THE
CEILING PLENUM.
CEILINGS 9'-0", ROOF DECK 12'-0":
A 3'-0" PLENUM. SEE SECTION, M-601.
DIFFUSER TYPES AND CFM PER M-501.
FLEX DUCT 8"ø, 6'-0" MAX, TO EACH
DIFFUSER FROM A TAP W/ DAMPER.
HOOD EXHAUST (GD): 18"ø WELDED 16 GA
BLACK STEEL, SLOPE TO HOOD, C.O. AT
THE ELBOW, 18" CLEAR OF COMBUSTIBLES,
UP TO EF-1 (NFPA 96, IMC 506).
KITCHEN / HALL WALL IS 1-HR RATED:
UL 555 FIRE DAMPER AT EACH DUCT
PENETRATION (IMC 607.5.1). NO DAMPER
OF ANY KIND IN THE GREASE DUCT.
MAU-1 INTERLOCKED WITH EF-1.
AIR BALANCE: SUPPLY 2,650 + MAKE-UP
2,000 · EXHAUST 2,400 + 225:
BUILDING SLIGHTLY POSITIVE.
RTU-1: 3,000 CFM, 1.0" ESP.



COUNTTOOLING — SAMPLE PLAN

PROJECT MAIN ST RESTAURANT

SHEET



SCALE

DATE

1/8" = 1'-0"

07/31/26

MECHANICAL PLAN

DRAWN: CT

M-101

EQUIPMENT SCHEDULE

TAG	DESCRIPTION	CFM	ESP	COOLING	ELEC	WEIGHT
RTU-1	ROOFTOP UNIT, GAS HEAT / DX COOL	3,000	1.0"	7.5 TON	208V 3Φ	900 LB
EF-1	HOOD EXHAUST FAN, UPBLAST	2,400	1.25"	-	208V 1Φ	150 LB
MAU-1	MAKE-UP AIR UNIT, TEMPERED	2,000	0.5"	-	208V 3Φ	650 LB
EF-2	RESTROOM EXHAUST FAN	225	0.4"	-	120V	40 LB

DIFFUSER AND GRILLE SCHEDULE

TAG	DESCRIPTION	NECK	CFM	ROOMS
SD-1	SUPPLY DIFFUSER, 24X24 LAY-IN, 4-WAY	8"ø	150	DINING, BAR, DISH
SD-2	SUPPLY DIFFUSER, 12X12 SURFACE	6"ø	100	HALL, STORAGE
SD-3	SUPPLY DIFFUSER, 24X24 LAY-IN, 4-WAY	10"ø	200	KITCHEN
RG-1	RETURN GRILLE, 24X24 LAY-IN, PLENUM	-	-	DINING, KITCHEN
EG-1	EXHAUST GRILLE, 8X8	6"ø	75	RESTROOMS, MOP
MA-1	MAKE-UP AIR REGISTER, 24X24	20X16	2000	KITCHEN

ROOM AIR SCHEDULE

ROOM	AREA SQ FT	SUPPLY CFM	EXHAUST CFM	SERVED BY
DINING 100	1,104	1,200	-	RTU-1
BAR 101	261	300	-	RTU-1
MEN 102	148	-	75	EF-2
WOMEN 103	137	-	75	EF-2
MOP 104	116	-	75	EF-2
KITCHEN 105	459	800	2,400	RTU-1 / EF-1 / MAU-1
DISH 106	126	150	-	RTU-1
HALL 107	117	100	-	RTU-1
STORAGE 108	216	100	-	RTU-1

SCHEDULE NOTES

1. SUPPLY CFM PER ROOM FROM THE COOLING LOAD AND ASHRAE 62.1 VENTILATION; USE THESE, NOT A RULE OF THUMB.
2. RTU-1 SUPPLIES 2,650 CFM OF ITS 3,000; THE REST IS FUTURE.
3. KITCHEN: HOOD EXHAUST 2,400 CFM (EF-1), MAKE-UP 2,000 (MAU-1), SUPPLY 800 (RTU-1).
4. RESTROOMS EXHAUST ONLY; AIR IS DRAWN FROM THE HALL UNDER THE DOORS (IMC 403, TABLE 403.3.1.1).
5. FLEX DUCT 6'-0" MAX PER DROP.

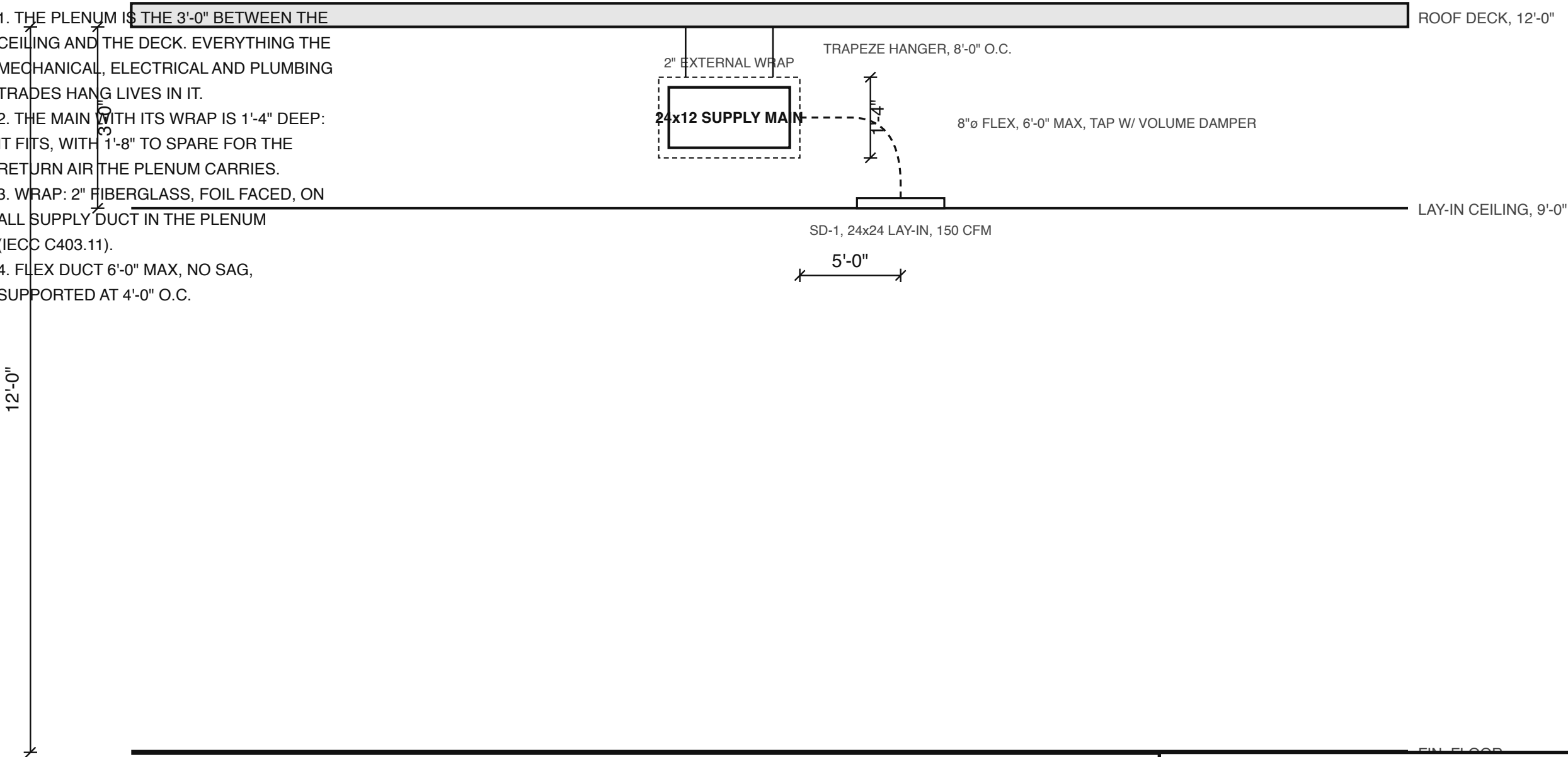


1 SECTION THROUGH THE DINING ROOM CEILING

SCALE: 1/2" = 1'-0"

SECTION NOTES

- 1. THE PLENUM IS THE 3'-0" BETWEEN THE CEILING AND THE DECK. EVERYTHING THE MECHANICAL, ELECTRICAL AND PLUMBING TRADES HANG LIVES IN IT.
- 2. THE MAIN WITH ITS WRAP IS 1'-4" DEEP: IT FITS, WITH 1'-8" TO SPARE FOR THE RETURN AIR THE PLENUM CARRIES.
- 3. WRAP: 2" FIBERGLASS, FOIL FACED, ON ALL SUPPLY DUCT IN THE PLENUM (IECC C403.11).
- 4. FLEX DUCT 6'-0" MAX, NO SAG, SUPPORTED AT 4'-0" O.C.



COUNTTOOLING — SAMPLE PLAN

PROJECT MAIN ST RESTAURANT

SHEET

SCALE
1/2" = 1'-0"

DATE
07/31/26

SECTIONS

DRAWN: CT

M-601